

Response to the Editor and Reviewer 1

Manuscript QSS-2026-0077 (Major Revision)

Citation-Network Cohesion and Reciprocal Citation Cliques Across Journal-Impact Groups: A Matched Author Analysis

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Dear Dr. Larivière,

Thank you for inviting a major revision of manuscript QSS-2026-0077. We also thank Reviewer 1 for the detailed assessment and for identifying the need for a clearer and more rigorously justified contribution. We rebuilt the affected analysis before revising the prose. The reviewer comments are reproduced below in their original order, followed by point-by-point responses. Section, table, and figure references correspond to the final compiled revision.

Comment 1: Overall assessment

This paper investigates whether authors publishing in low-impact journals exhibit denser and more reciprocated citation patterns compared to their counterparts in high-impact venues. The topic is of undoubted interest, but in its current form, it prioritises technical virtuosity over substantive contribution. For a journal like QSS, which values methodological rigour alongside clarity, a major revision is essential. The authors must reframe their work for a broader audience, clarify their analytical pipeline, and strengthen the connection between their findings and the scholarly literature. However, given the number and severity of the issues identified below, I regret to recommend rejection in its current form.

Response. We agree that the earlier draft required a substantive rather than cosmetic revision. We retitled the manuscript from *Citation Cliques in Low Impact Journals* to *Citation-Network Cohesion and Reciprocal Citation Cliques Across Journal-Impact Groups: A Matched Author Analysis* to identify the defined structural analysis while avoiding an unsupported allegation. The paper is now framed as an associational matched comparison, not an assessment of scientific merit or evidence of intent. We retain the motivating directional comparison as an explicit hypothesis about higher primary cohesion and outgoing concentration among Cases, while distinguishing positive mean differences from zero-inflated median shifts. The full-cohort clique analysis gives that hypothesis a matched structural test: primary-rule membership was 10.96% for Cases and 3.82% for Controls, a paired difference of 7.14 percentage points (exact $p < 0.001$). The analytical sequence is shown in a reader-facing design figure and presented as Data and Cohort, Matching, Network Construction and Definitions, Paired Inference, Anomaly Screen, Results, Discussion, Limitations, and Conclusion. The feature set and claims were reduced, affected results were regenerated from one canonical workflow, and the Discussion now interprets the findings in relation to the author- and journal-level literature.

Location. Title, Abstract, Figure 1, Discussion, and Conclusion.

Comment 2: Clarity and Presentation: terminology

The primary obstacle is the paper’s prose. Specialised terminology is deployed without adequate explanation, and crucial concepts are often introduced before being defined. This disrupts the narrative flow and assumes a level of technical familiarity that may not align with the journal’s readership. I suggest to the authors an effort to translate the methodological procedures into clear language.

Response. We rewrote the Methods so that the unit of analysis, data identifiers, matching variable, graph populations, and notation are defined before the analyses that use them. The revision expands DOI, ISSN, and HHI at first use; defines h_5 operationally; distinguishes the outgoing ego, matched-author, and work-reference populations; and gives both formal and plain-language definitions of the directed graph and its binary projection. It also gives plain-language explanations of Isolation Forest and assortativity and states how positive effect sizes should be read. The main text now gives the reader-facing explanation first, while the appendix retains the exact implementation details needed for reproducibility.

Location. Data and Cohort through Anomaly Screen; Table 2.

Comment 3: Clarity and Presentation: metric definitions and burst intensity

*The presentation of metrics in Table 5 exemplifies this problem. Several measures lack clear, self-contained definitions, and the bibliographic references provided are not illuminating: they point to general theoretical frameworks without specifying how these translate into the precise operationalisations actually adopted in the study. The Normalised burst intensity, for instance, is defined as $\max_j w_{ij}/(\text{in_strength} + 1)$ and described as the largest single-peer citation weight normalised by total incoming volume, yet the notation (w_{ij} , **in_strength**) and the connection to Kleinberg’s burst-detection methods remain under-explained. What is missing is an explicit link between the inspiring principle and its concrete implementation, which for a journal like QSS should be made transparent.*

Response. The former Table 5 has been replaced by Table 2. It now states each formula, denominator, analytic population, and interpretation. Its notes define every symbol used in the table, explain the HHI scale, distinguish structural zeros from unavailable denominators, and specify missing-value handling. The surrounding Methods also gives the exact reference-deduplication and fractional-weight construction that supplies those formulas.

The former normalised burst-intensity variable has been removed. It is replaced by the study-defined maximum annual dyadic surge share, whose complete formula compares consecutive dyad-years and divides the largest positive increment by the author’s total identified outgoing weight. Absent dyad-years are treated as zero for this calculation, authors without identified outgoing weight remain missing, and the result is bounded by $[0, 1]$. The manuscript explicitly states that this is a five-year scalar summary, not a trajectory model or an implementation of Kleinberg’s state-transition burst method; the Kleinberg attribution has therefore been removed. References now indicate conceptual provenance; each study-specific operationalization is stated in full rather than delegated to those sources.

Location. Network Construction and Definitions, Paired Inference, and Limitations; Table 2.

Comment 4: Clarity and Presentation: organization, K-means, and literature

Furthermore, the paper conflates methods, analysis, and results, often presenting findings before the underlying techniques have been introduced. The three central groups from the k-means clustering, for instance, appear without any description of the clustering procedure. Restructuring the narrative to follow a logical sequence would greatly improve comprehension. Importantly, the discussion fails to connect the empirical results back to the literature review, making the paper feel like an isolated technical exercise rather than a contribution to the ongoing scholarly conversation.

Response. All essential definitions and analytical decisions now precede Results, which begins only after the Data, Matching, Network, Paired Inference, and Anomaly Screen sections. Figure 1 summarizes that sequence. K-means and the former three-cluster result have been removed because they are no longer part of the revised study.

The Discussion now connects the regenerated findings, not only the study design, to prior work. It distinguishes outgoing HHI from Evdaimon et al.’s incoming-citer concentration indicator, uses the tie-heavy zero-median findings to limit interpretation of cohesion differences, contrasts lower journal endogamy and higher surge share with Heneberg’s journal-level kinetics, and distinguishes the author-level matched clique result from the journal-level anomalous-group detection studied by Kojaku et al. It also discusses non-adversarial alternatives and states why these studies are conceptually informative but not numerically comparable.

Location. Methods through Results, and Discussion.

Comment 5: Methodological Rigour and Transparency: matching

The technical execution, while ambitious, contains critical problems. The matching methodology suffers from a logical inconsistency: the bucketing formula $b = \lfloor h_5/3 \rfloor$ permits a five-point h_5 -index difference, contradicting the intended three-point limit. Additionally, no sorting order is specified for the greedy one-pass algorithm. These fundamental issues must be resolved to ensure methodological transparency and credibility.

Response. We replaced the bucket rule with the explicit inclusive eligibility condition $|h_{5,\text{Case}} - h_{5,\text{Control}}| \leq 3$. The revised Methods defines h_5 and specifies the complete greedy order: subject, ascending absolute h_5 difference, Case ORCID, and Control ORCID. Selection is without replacement within subject. The full-cohort h_5 table admits every non-empty ORCID but retains the positive- h_5 rule, so otherwise tier-eligible zero- h_5 profiles do not enter candidate generation.

An immutable read-only audit recomputed all candidates and the greedy traversal from the generated full-cohort author profiles and positive h_5 values. It reproduced the materialized 102,626-pair order, membership, and pair-set hash exactly. It does not claim to represent profiles with $h_5 = 0$. The maximum observed absolute difference is 2, with 0 caliper violations and no reused author–subject keys. Matching balance is reported by subject, and the four primary comparisons are repeated for exact- h_5 pairs.

Location. Matching and sensitivity results; Tables 1 and 5.

Comment 6: Methodological Rigour and Transparency: outlier framework

The outlier-detection framework also requires substantial justification. The rationale for feature selection, hand-tuned weighting, and the four-sigma cutoff is not adequately explained. The use of feature weights in an Isolation Forest model is mathematically redundant, as the algorithm is not designed for such inputs, and the four-sigma threshold implies a normality assumption that is inappropriate for skewed bibliometric data. The authors should provide a clearer justification for these decisions, report sensitivity analyses to test the stability of their findings, and clarify whether the Isolation Forest and Cohesion Composite Score capture redundant information.

Response. The revised specification uses an explicit structural rule rather than an optimized feature search. It retains five author-edge measures corresponding to self-reference, prior-collaborator citation, mutual exchange, neighborhood closure, and recipient concentration. Journal endogamy is excluded because it is a work-reference measure, balance because it is a signed internal-flow position unavailable without internal incident weight, and surge share because it is temporal. We present this as a transparent revision choice, not as an originally prespecified or independently validated optimum.

No additional feature weights are applied. The four-sigma rule and former Cohesion Composite Score have been removed from the revised canonical analysis and manuscript. Controls define subject-specific median/IQR scaling, model fits, and score cutoffs; a nonfinite or zero Control IQR is replaced by 1 solely to avoid division by zero. The declared 99th Control percentile is a stringent empirical-tail operating point that makes no Gaussian assumption and is not claimed to be uniquely optimal. The library's `contamination=auto` decision cutoff is not used; the reported criterion is computed directly from negated `score_samples`.

Sensitivity analyses evaluate 97.5%, 99%, and 99.5% Control percentiles across ten fixed seeds and report leave-one-feature-out fits. Because all four cohesion-restriction measures also enter the detector, the manuscript explicitly treats their intersection as an overlapping guardrail rather than independent corroboration. The complete overlap table and rank association quantify that dependence. The two screen rules agreed for 2,042 rows; 3,930 passed only the first rule and 14 only the second. Their scores were positively related ($\rho = 0.421$, $p < 0.001$). Removing one input at a time changed little: flag overlap ranged from 0.931 to 0.995, and Case/Control enrichment ranged from 14.83 to 15.58.

Location. Anomaly Screen and anomaly results; Tables 6, 7, 8, and 12.

Comment 7: Inconsistencies and Omissions

Several inconsistencies and omissions further weaken the analysis. The manuscript does not specify the underlying statistical test applied to the matched pairs, a critical omission given the non-normal nature of the data. The claim of a widening gap in “citation trajectories” is purely descriptive and no time series data are reported to support this assertion. On page 9, the authors stated that “normalised burst intensity” was dropped, leaving 13 features for the analysis. However, on page 13, they report that “12 out of 14 tested behavioral metrics remain significant”. The authors should clarify whether the missing feature was reintegrated for the univariate statistical tests (by dropping the incomplete pairs) or if this is a typographical error.

Response. The revised Methods specifies a two-sided paired Wilcoxon signed-rank test for every outcome because several paired-difference distributions are bounded and tie-heavy. Zero differences are omitted from the signed ranks, unavailable pairs are deleted separately for each outcome, and Wilcoxon p -values are Benjamini–Hochberg adjusted within the four-outcome primary and secondary families. Each table reports its own complete-pair count, median paired difference, mean paired difference, both paired-bootstrap intervals, matched rank-biserial effect, and sign-flip randomization check. Because the measures are zero-inflated, mean paired differences and their intervals are descriptive complements; they do not replace the median-based nonparametric summary. The four primary mean paired differences were Coauthor-citation rate 0.048, 95% bootstrap CI [0.047, 0.050]; Reciprocity 0.096, 95% bootstrap CI [0.092, 0.100]; Local clustering 0.081, 95% bootstrap CI [0.079, 0.084]; Outgoing HHI 0.097, 95% bootstrap CI [0.095, 0.099]. 3 median differences were zero. Outgoing HHI had a median paired difference of 0.036, 95% bootstrap CI [0.035, 0.037].

The earlier draft did not establish whether the 13-versus-14 discrepancy arose because normalised burst intensity was reintegrated for per-metric tests or because one count was typographical. We therefore withdraw both statements rather than assign a retrospective explanation. The rebuilt analysis reports eight outcomes in two declared families, separately from five screen inputs. The former burst variable is replaced by maximum annual dyadic surge share, with 75,526 complete pairs in its secondary comparison.

We removed the earlier widening-gap claim. The year field supports coauthor timing and the scalar surge-share construction, but the revision estimates neither tier-specific trajectories nor a tier-by-year interaction. The Limitations now states this explicitly.

Location. Paired Inference, matched results, and Limitations; Tables 3, 4, and 5.

Comment 8: Network Characterisation

The central qualitative claim of a “hub-and-spoke” topology is presented without any formal network definition, graph-theoretic metrics, or quantitative thresholds. The role of Louvain community detection is also unclear. From Table 5 it is unclear if the adjustments for directed networks are adopted in the computation of network indicators such as eigenvector. Moreover, it is unclear on which graphs some indicators are calculated (in table 5 the manuscript refers to “undirected projection” and “undirected citation” graphs without specifications. To move from observation to verifiable evidence, the authors must provide the analytical basis for their network characterisation.

Response. We replaced the former qualitative hub-and-spoke claim and unused Louvain result with a formal clique analysis. On each subject-specific matched-author graph, a clique is a maximal clique in the positive undirected projection with at least four nodes. We report directed density, defined as positive directed dyads divided by $k(k - 1)$, and weighted reciprocity, defined as the sum of pairwise minimum weights divided by the sum of pairwise maximum weights. The primary rule requires density at least 0.75 and reciprocity at least 0.50; minimum sizes 3, 4, and 5 and reciprocity thresholds 0.25, 0.50, and 0.75 are reported as sensitivity analyses.

For inference, we convert overlapping qualifying cliques to one binary membership value per author–subject and compare Case and Control membership within the original matched pairs. The exact two-sided pair-label null conditions on discordant pairs and gives either member equal probability of carrying the membership label. We report Case and Control rates, their paired difference, 10678 Case-only and 3351 Control-only pairs, the exact p -value, and subject-level direction for every threshold

sensitivity. This directly matches the sampling design and avoids counting one author repeatedly because cliques overlap. The anomaly flag is not used to define the clique population. This provides the requested graph definition, quantitative thresholds, directed adjustments, and matched inference without treating a structural clique as evidence of intent.

The revision separates the journal and author networks. Eigenfactor is a fixed journal-level exposure used before author matching, not an author centrality. The Methods now specifies citing-to-cited orientation, journal self-citation removal, outgoing-column normalization, 0.85 damping, article-count redistribution of dangling mass and teleportation, and within-subject normalization. The author analysis therefore treats the venue score as a fixed exposure and estimates downstream structural differences without conflating journal centrality with author-network measures.

For the author analysis, the Methods formally defines the loopless directed weighted graph G_s and its binary undirected projection U_s . It identifies the population, direction, weight, denominator, and missingness rule for every measure. Direction is discarded only for U_s clustering and clique membership; reciprocity, balance, tier mixing, and the clique density measure retain direction and fractional weights. The tier-mixing label-swap p -value tests the within-tier fractional-weight share, while weighted assortativity is reported separately as a descriptive statistic. The new clique tables and Figure 5 provide the analytical and visual basis for the network characterization.

Location. Data and Cohort, Network Construction, Anomaly Screen, and network results; Figure 5; Tables 2, 9, 10, and 11.

Comment 9: Conclusion

In summary, this manuscript contains interesting ideas, but the priority for revision must be clarity, rigorous methodological justification, and a stronger connection to the literature. The current draft reads as a proof of technical capability. I suggest that the authors transform it into an accessible contribution to quantitative science studies.

Response. We agree with this priority. The revised paper leads with the substantive matched comparison, defines its technical terms and populations before use, limits the analysis to auditable operationalizations, reports uncertainty and sensitivity rather than a proliferation of exploratory outputs, and interprets the regenerated findings against the relevant literature. The primary tables now show both median and mean paired differences, making the zero-inflated, tie-heavy outcome distributions explicit. The conclusion states the matched result directly: lower-impact-group authors were more likely than matched higher-impact-group authors to belong to formally defined reciprocal citation cliques. The wording remains associational and does not infer coordination, intent, or misconduct from structural evidence alone.

Location. Abstract, Discussion, Limitations, and Conclusion.

Additional corrections incorporated in the revision

Beyond the issues named above, the rebuild makes subject part of every author and edge key; deduplicates work references before author expansion; counts complete author teams in fractional denominators while requiring identified graph endpoints; separates ego, induced-graph, and work-reference populations; preserves unavailable values; and uses the same final flag for enrichment and

figures. All manuscript values, tables, and figures are generated by one declared canonical analysis; earlier estimates are not used.

Location. Author–subject cohort through Paired Inference, Anomaly Screen, Results, and Data and Code Availability.